Akshay Choudhary

(+91) 8827406511 akshaychoudhary2040@gmail.com linkedin github

Education

National Institute of Technology, Rourkela **Aug 2023 – Present**
Bachelor of Technology in Biotechnology — CGPA: 8.59 *Rourkela, Odisha*

Experience

Research Intern **May 2026 – Present**
IIT Kharagpur (Grisha Research Internship Program)

- Designing and evaluating deep learning experiments involving representation learning, model benchmarking, and statistical performance analysis.
- Reproduced and evaluated research-grade neural architectures using PyTorch and scientific datasets.
- Designed reproducible experimentation pipelines for model training, hyperparameter tuning, and performance evaluation.
- Investigating representation learning techniques for improving predictive performance in scientific machine learning tasks.

Open Source Contributions

PyCQA/Pylint Contributor | *Python, Static Analysis, Open Source* **Github**

* Contributed fixes to a widely used Python static analysis tool improving linting accuracy and Python compatibility.
* Merged multiple pull requests addressing static analysis correctness, inference handling, and Python compatibility issues.
* Investigated and resolved edge cases involving type inference, AST analysis, and static code checking.
* Collaborated with maintainers through code reviews, issue discussions, and iterative patch development.

Projects

MiniML - Machine Learning Library | *Python, NumPy, PyPI, GitHub, CI/CD* **Github**

* Developed and published a machine learning library implementing Linear Regression, KNN, K-Means, DBSCAN and other core ML algorithms.
* Designed Scikit-learn-inspired APIs supporting model training, preprocessing, evaluation, and inference workflows.
* Built modular and extensible architecture with automated testing, package versioning, and CI/CD through GitHub Actions.

AI-Powered Meeting Minutes Generator | *Pegasus, FastAPI, Hugging Face, React, Vercel* **Github**

* Built an end-to-end LLM-powered system for automatic meeting transcription and abstractive summarization.
* Developed data preprocessing pipelines for transcript cleaning, tokenization, and conversational normalization.
* Fine-tuned and evaluated Pegasus-based summarization workflows, improving summary quality measured through ROUGE evaluation.
* Designed scalable inference APIs and integrated frontend workflows for real-time summary generation.

Deep Learning for Enzyme Kinetics Prediction | *PyTorch, Deep Learning, GNNs, Google Colab* **Github**

* Implemented the DLKcat architecture for enzyme turnover number (kcat) prediction using PyTorch.
* Integrated protein sequence embeddings and molecular graph representations for regression modeling.
* Reproduced baseline performance with R^2 of 0.46, RMSE of 3.67, and MAE of 2.72.
* Developed reproducible training and evaluation workflows for scientific machine learning experiments.

Technical Skills

Programming: Python, SQL, C++, C, JavaScript
Machine Learning & Deep Learning: PyTorch, Neural Networks, Deep Learning, Scikit-learn, Hugging Face, NLP
Backend: FastAPI, REST APIs, Flask
Data Science: Pandas, NumPy, Statistics, Matplotlib, Feature Engineering, Model Evaluation, Cross Validation, EDA, Visualization
Vector Databases: FAISS, Pinecone, ChromaDB
Developer Tools: Git, GitHub, Docker, Jupyter Notebook, Streamlit

Achievements

· Top 10 Finalist – Teach-a-thon Hackathon (500+ teams)
· Codeforces (Max Rating: 922) and CodeChef 3-Star (Max Rating: 1346)
· Solved 300+ DSA problems
· Organized HackOdisha 4.0, a 36-hour national hackathon involving 100+ participating teams.